

NANGRONG, Thailand

WMO: 484360

Lat: 14.631N

Lon: 102.721E

Elev: 182

StdP: 99.16

Time Zone: 7.00 (E07)

Period: 01-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) 15.0	(c) 16.3	(d) 10.4	(e) 8.0	(f) 19.7	(g) 11.9	(h) 8.9	(i) 19.8	(j) 6.4	(k) 24.9	(l) 5.5	(m) 24.0	(n) 0.6	(o) 10	(p) 0.313

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
4	10.9	38.1	25.4	36.9	25.4	35.9	25.5	27.4	33.3	27.1	32.9	26.9	32.5	1.6	120

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
26.2	22.0	29.2	25.8	21.5	28.9	25.6	21.3	28.8	88.2	33.5	86.9	33.0	85.7	32.7	33.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years						
Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max					
(a)	(b)	(c)		(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
5.7	4.6	3.9	DB	12.5	39.6	1.3	1.2	11.6	40.4	10.9	41.1	10.2	41.8	9.2	42.6	
			WB	11.2	28.3	1.3	0.5	10.3	28.6	9.6	28.9	8.8	29.2	7.9	29.6	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	27.9	24.6	26.9	29.2	30.4	29.8	29.6	29.0	28.8	28.1	27.4	26.1	24.5
	DBStd	2.65	2.55	2.62	2.36	1.81	1.43	1.08	1.18	1.04	1.08	1.19	2.04	2.50
	HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD18.3	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDD10.0	6521	452	473	594	611	614	588	589	582	544	539	484	450
	CDD18.3	3480	193	240	336	361	356	338	331	324	294	281	234	192
	CDH23.3	37353	1726	2609	4088	4502	4131	3958	3686	3486	2887	2616	2055	1609
	CDH26.7	16683	719	1317	2213	2425	1931	1791	1568	1402	1035	899	774	608
Wind	WSAvg	1.0	1.0	0.9	0.9	0.8	0.8	1.0	1.1	1.1	0.8	1.1	1.4	1.4
Precipitation	PrecAvg	1180	6	19	47	76	151	134	137	197	243	138	30	4
	PrecMax	1441	27	72	149	155	223	277	242	355	347	293	113	21
	PrecMin	920	0	0	2	16	63	50	59	91	118	78	0	0
	PrecStd	170	8	22	41	39	43	69	51	56	63	53	33	6
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	34.9	37.4	39.2	40.2	38.0	36.6	36.0	35.1	34.3	33.8	34.7	34.3
		MCWB	23.4	24.1	24.3	25.9	26.8	26.2	26.1	26.4	26.6	26.0	25.3	24.1
	2%	DB	33.5	36.1	37.9	38.4	36.4	35.4	34.7	34.0	33.3	32.6	33.1	32.9
		MCWB	23.4	24.1	24.6	25.9	26.6	26.3	26.1	26.2	26.6	26.0	25.0	24.0
	5%	DB	32.2	34.9	36.7	37.0	35.2	34.3	33.7	33.2	32.4	31.7	32.0	31.6
		MCWB	22.9	24.0	24.8	25.9	26.6	26.2	26.1	26.1	26.4	25.7	24.8	23.4
	10%	DB	30.8	33.5	35.4	35.6	33.9	33.4	32.7	32.2	31.5	30.9	30.8	30.2
		MCWB	22.4	23.8	24.8	25.7	26.5	26.2	25.9	26.0	26.1	25.5	24.4	22.9
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	25.6	26.4	26.8	27.5	28.0	27.5	27.3	27.3	27.6	27.1	26.8	25.7
		MCDB	31.7	33.8	34.6	35.9	34.2	33.3	32.9	32.7	32.2	31.3	31.6	31.5
	2%	WB	24.7	25.6	26.2	26.9	27.5	27.1	26.7	27.0	27.1	26.7	26.2	24.9
		MCDB	30.8	32.6	33.8	34.8	33.6	32.8	32.1	32.2	31.8	30.8	30.9	30.6
	5%	WB	24.0	25.0	25.7	26.6	27.1	26.7	26.5	26.6	26.7	26.3	25.7	24.2
		MCDB	29.5	32.0	33.1	34.1	33.1	32.2	31.8	31.5	31.1	30.2	30.0	29.6
	10%	WB	23.3	24.4	25.4	26.2	26.8	26.5	26.2	26.3	26.3	26.0	25.2	23.6
		MCDB	28.7	31.0	32.6	33.4	32.3	31.7	31.3	30.9	30.4	29.7	29.0	28.6
Mean Daily Temperature Range	5% DB	MDBR	12.5	12.9	11.9	10.9	8.8	8.2	7.9	7.7	7.1	7.2	9.1	11.0
		MCDBR	13.5	14.1	13.8	12.6	10.4	9.4	9.1	8.7	8.3	8.1	10.3	11.9
		MCWBR	4.3	3.8	2.9	2.8	2.6	2.3	2.5	2.6	2.7	2.6	3.3	3.8
	5% WB	MCDBR	11.3	12.1	11.6	11.2	9.2	8.5	8.3	8.2	7.7	7.4	8.8	10.6
MCWBR		3.8	3.6	3.0	3.0	2.8	2.5	2.7	2.7	2.8	2.7	3.0	3.8	
Clear-Sky Solar Irradiance	taub		0.502	0.611	0.650	0.589	0.496	0.466	0.457	0.460	0.491	0.526	0.480	0.468
	taud		2.026	1.774	1.726	1.908	2.212	2.300	2.320	2.297	2.200	2.068	2.178	2.159
	Ebn at Noon		790	721	707	751	811	827	837	843	821	781	806	810
	Edh at Noon		168	225	241	201	146	132	130	134	148	165	143	144
All-Sky Solar Radiation	RadAvg		4.98	5.23	5.34	5.68	5.54	5.26	4.98	4.84	4.65	4.78	4.88	4.84
	RadStd		0.30	0.22	0.23	0.26	0.32	0.20	0.29	0.28	0.26	0.35	0.28	0.17

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (19 neighbors)	+0.31	N/A	N/A	N/A	+0.26	+0.22	N/A	N/A	+131	+131

Nomenclature: See separate page